

Agentic RFP Evaluation and Supplier Ranking

Project submission report | 27 September 2026

This project evaluates supplier proposals for a fictional university procurement request. It reads each PDF, asks a language model for criterion scores with document evidence, checks the model response, and then uses fixed Python rules to calculate scores and rank suppliers. A completed four-supplier run, validation example, source code, and fictional proposals are included.

What the application delivers

Requirement	Implemented behavior
Criteria	SQLite stores criterion names, descriptions, weights, maximum scores, and active status. The editor enforces a 100% active total.
Supplier input	Streamlit accepts multiple PDFs, supplier names, dates, and experience ratings.
AI evaluation	Gemini or Ollama Cloud receives extracted text and current criteria; it returns JSON scores, reasons, evidence, risks, and a summary.
Validation	Missing or invalid scores become zero, out-of-range scores are clipped, and corrections appear as warnings.
Ranking and records	Python computes all measures and tie breaks; SQLite saves a full batch under one run ID; JSON export is available.

System flow

The orchestrator loads the active criteria immediately before a batch. For each supplier, it extracts selectable PDF text, builds a prompt from the current criteria, calls the selected model, and normalizes the JSON response. After all suppliers are assessed, the scoring module finds criterion benchmarks, computes the peer measures, sorts the suppliers, and saves the completed result. The interface shows the leaderboard, criterion details, evidence, warnings, and a download button.

Scoring and ranking rules

Measure	Calculation
Absolute score	Sum of (criterion score / maximum score) x weight
Benchmark and gap	Highest valid criterion score in the batch; supplier score minus benchmark
Relative performance	Supplier score / benchmark x 100; use 100% when the benchmark is zero
Peer Performance Index	Weighted average of criterion relative percentages
Final rank	Higher PPI, earlier date, higher experience rating, supplier name A to Z

Completed Gemini evaluation

Four fictional two-page proposals were evaluated with Gemini 2.5 Flash. The model produced criterion-level judgments; Python calculated the following final values.

Rank	Supplier	Absolute score	PPI
1	Apex Systems	85.0	94.4445
2	NexaWorks	82.0	91.1111
3	Orbit Digital	67.0	74.4445
4	BrightPath Tech	47.0	52.2222

Apex Systems ranked first with strong architecture and security detail. NexaWorks followed with a strong implementation plan and support offer. Orbit Digital had strong references but less integration detail. BrightPath Tech had the lowest price but limited security detail. The model returned a malformed risks list for BrightPath; the validator recorded a warning and retained the valid criterion scores.

Validation and testing

The separate validation example includes an out-of-range criterion score that is corrected to zero. Four automated tests pass. They cover missing and malformed results, score clipping, weighted scores and peer measures, all tie breaks, a zero benchmark, and complete SQLite persistence. The local Streamlit interface was opened and checked for the criteria view, evaluation form, sample leaderboard, scorecards, and export control.

Project files and demonstration

The repository contains the Streamlit application, Python modules, requirements file, database seed script, four proposal PDFs, the completed JSON export, the validation example, and tests. To start locally, install requirements, run the seed script, and launch streamlit run streamlit_app.py. The Sample run tab displays the completed evaluation without a key. For a new batch, upload proposal PDFs and enter a Gemini or Ollama Cloud key in the AI connection panel.

Technical references

Google AI for Developers, Gemini generateContent API: <https://ai.google.dev/api/generate-content>

Streamlit, Deploy your app on Community Cloud: <https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app/deploy>

Streamlit, Secrets management: <https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app/secrets-management>

Application screenshots

Completed four-supplier Gemini evaluation in the Sample run tab.

RFP Supplier Evaluation

Compare supplier proposals using document evidence and transparent ranking rules.

Evaluate Criteria Saved runs Sample run

Example

Live Gemini evaluation

Both examples use the four fictional supplier proposals.

Leaderboard

Rank	Supplier	Absolute score / 100	PPI / 100	Submitted	Experience / 10
1	Apex Systems	85	94.4445	2026-09-15	8.5
2	NexaWorks	82	91.1111	2026-09-16	8
3	Orbit Digital	67	74.4445	2026-09-13	9.5
4	BrightPath Tech	47	52.2222	2026-09-14	4

Order: higher PPI, earlier submission, higher experience, then supplier name A to Z.

Active criteria and weights loaded from SQLite.

RFP Supplier Evaluation

Compare supplier proposals using document evidence and transparent ranking rules.

Evaluate Criteria Saved runs Sample run

Evaluation criteria

criterion_id	name	description	weight	max_score	Active
1	Technical Capability	Architecture, integrations, scalability, technical fit	30	10	☒
2	Implementation Plan	Timeline, milestones, staffing, risk plan	20	10	☒
3	Commercial Value	Pricing clarity, total cost, assumptions	20	10	☒
4	Security & Compliance	Controls, certifications, privacy, auditability	20	10	☒
5	Support & Experience	Support model, similar projects, references	10	10	☒

Current active weight: 100%

Save criteria